NETFLIX USER ANALYTICS DASHBOARD REPORT (power BI)

Abstract

The objective of this project is to analyze Netflix user behavior using a data analytics dashboard. The dashboard provides insights into user demographics, revenue trends, device usage, payment methods, content preferences, and subscription types. By transforming raw data into meaningful visualizations, the dashboard helps stakeholders understand patterns in user engagement and platform performance. This project demonstrates how data analytics tools can convert large datasets into actionable insights. The dashboard allows decision-makers to monitor key performance indicators such as total users, total countries, ratings, and total revenue. Through various visualizations including charts, maps, and KPI cards, the project highlights important trends in streaming behavior and user preferences.

1. Introduction

The rapid growth of digital streaming platforms has generated large volumes of user interaction data. Analyzing this data helps organizations understand customer behavior, optimize content strategies, and improve revenue performance. This project focuses on analyzing Netflix user behavior using a visual analytics dashboard. The dashboard provides insights into user demographics, revenue generation, subscription types,
device usage, payment methods, and content preferences. By transforming raw data into visual reports, decision makers can quickly identify trends and patterns.

2. Objectives of the Project

- To analyze Netflix user behavior using data visualization.
- To understand the distribution of users across countries.
- To analyze revenue trends generated by subscriptions.
- To evaluate device usage patterns among users.
- To analyze content interaction based on age groups.
- To identify popular genres preferred by users.
- To create an interactive dashboard for business decision making.

3. Dataset Description

The dataset used in this project contains information about Netflix users and their streaming behavior.
The data was structured in tabular format and included several attributes related to user activity and subscription details.

Important columns in the dataset include:

- **User ID** – Unique identifier assigned to each user.
- **Gender –** Indicates whether the user is male, female, or other.
- **Age** – Represents the age group of the user.
- **Country –** Shows the geographical location of the user**.**
- **Device Used** – Indicates the device used for streaming (Laptop, Tablet, Smart TV, Mobile).
- **Payment Mode** – Shows the payment method such as PayPal, UPI, Credit Card, or Debit Card.
- **Subscription Type** – Indicates the plan chosen by the user (Basic, Standard, Premium).
- **Favorite Genre –** Represents the genre most watched by the user.
- **Revenue** – Shows the revenue generated from subscriptions.

The dataset was cleaned and structured before building the dashboard to ensure accurate analysis.

4. Methodology

The project followed a structured data analytics workflow consisting of several stages.

Data Collection:

The dataset was collected from an online dataset source and stored in CSV format.

Data Cleaning:

Data preprocessing was performed to remove duplicates, handle missing values, and ensure data consistency. This step ensured the dataset was ready for analysis.

Data Transformation:

Using Excel or Power BI tools, the dataset was transformed and categorized to support visual analytics. Columns were formatted properly and new calculated measures were created for analysis.

Data Visualization:

Interactive visualizations were created to present insights effectively. Various charts such as line charts, bar charts, maps, and KPI cards were used to represent different aspects of the dataset.

Dashboard Design:

All visual components were integrated into a single dashboard interface that allows users to interact with filters and explore different insights.

5. Tools and Technologies Used

Excel / Power BI:

Used for data cleaning, transformation, and dashboard development.

Power Query:

Used for data preparation and transformation.

Data Visualization:

Charts such as line charts, bar charts, maps, and KPI cards were used to represent the insights.

6. Dashboard Overview

Key Performance Indicators shown in the dashboard:

- Total Users: 19,930
- Total Countries: 10
- Ratings: 5.00
- Total Revenue: 51.30K

Filters such as Gender and User ID allow users to interactively explore the dataset.

7. Detailed Analysis

Payment Mode Analysis:

PayPal and UPI show higher usage among customers compared to credit and debit cards.

Country Distribution:

Users are distributed across multiple countries including the USA, Canada, India, Germany, France, Spain, Brazil, and the UK.

Content Interaction Based on Age:
Middle-aged users demonstrate higher engagement with streaming content.

Device Usage:
Laptops are the most frequently used streaming devices followed by tablets and smart TVs.

Monthly Revenue by Genre:
Genres such as Documentary and Sci-Fi generate strong revenue contributions.

Popular Genres:
Drama, Romance, and Documentary content show high audience engagement.

Subscription Analysis:
Standard subscription plans contribute the highest revenue compared to Basic and Premium plans.

8. Key Insights

- The platform has over 4000 active users.
- Digital payment methods dominate subscription payments.
- Laptop streaming is the most common device usage.
- Middle-age groups interact with content more frequently.
- Story-driven genres are highly popular.

9. Conclusion

This project demonstrates how analytics dashboards can transform raw data into meaningful insights.
The Netflix dashboard provides valuable information about user behavior, revenue trends, and
content preferences.

Future enhancements can include predictive analytics, machine learning-based recommendations,
and real-time data monitoring to improve decision-making.